NT-1b Scalar Form Factors: Main Derivation

One-Loop Nonlocal Form Factors for a Real Scalar Field
with Non-Minimal Coupling

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Abstract

We derive the one-loop nonlocal form factors $h_C^{(0)}(x)$ and $h_R^{(0)}(x;\xi)$ for a **real scalar field** with arbitrary non-minimal coupling ξ on a closed Riemannian 4-manifold, expressed in the Weyl–R^2 basis $\{C^2, R^2\}$. Starting from the Codello–Zanusso (CZ) heat kernel form factors [1], we:

1. specialise the five universal CZ form factors to the scalar operator data ($\mathrm{tr}(\mathbf{1}) = 1$, $\Omega_{\mu\nu} = 0$, $\mathbf{U} = \xi R + m^2$);
2. convert to the Weyl basis via the Gauss–Bonnet identity;
3. verify all four mandatory local-limit benchmarks;
4. express the spectral action form factors $F_1^{(0)}$ and $F_2^{(0)}$ as explicit integrals over the spectral function ψ and its antiderivatives Ψ_1, Ψ_2;
5. analyse the ξ-dependence structure, limiting cases, and UV asymptotic behaviour.

All intermediate results are verified numerically using `scipy.integrate.quad` to machine precision ($< 10^{-14}$). This document parallels the NT-1 Dirac derivation and serves as input for the combined scalar + vector + Dirac form factor assembly in NT-1b.

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1 Setup and Notation

1.1 Operator data for the real scalar

We work on a closed Riemannian 4-manifold (M, g) with a real scalar field φ governed by the classical action

$$S[\varphi] = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{g}\,\{g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi + (m^2 + \xi R)\,\varphi^2\}\,, \tag{1}$$

where m is the mass and ξ is the non-minimal coupling parameter. The kinetic operator is

$$\Delta = -\Box + \xi R + m^2\,, \tag{2}$$

which is of Laplace type $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ with the following operator data (see [7] for complete derivation):

$$\boxed{\mathrm{tr}(\mathbf{1}) = 1\,, \qquad \Omega_{\mu\nu} = 0\,, \qquad E = -\xi R - m^2\,, \qquad \mathbf{U}_{\mathrm{CZ}} = -E = \xi R + m^2\,.} \tag{3}$$

The BV potential is $P_{\mathrm{BV}} = E + R/6 = -\tilde{\xi}R - m^2$ where we define

$$\tilde{\xi} \equiv \xi - \frac{1}{6} \tag{4}$$

as the deviation from conformal coupling.

1.2 One-loop effective action

The one-loop effective action for the scalar field is

$$\Gamma^{(1)} = +\frac{1}{2}\,\mathrm{Tr}\ln\Delta\,, \tag{5}$$

where the $+1/2$ reflects the bosonic nature of the field (contrast with -1 for Dirac fermions). At second order in curvature, the effective action takes the form

$$\Gamma^{(1)}\Big|_{\mathcal{R}^2} = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{g}\,[h_C^{(0)}(\Box/m^2)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + h_R^{(0)}(\Box/m^2;\xi)\,R^2] + \mathrm{E}_4\ \text{terms}\,, \tag{6}$$

where $h_C^{(0)}$ and $h_R^{(0)}$ are the nonlocal form factors to be computed, and E_4 denotes the topological Gauss–Bonnet term which is irrelevant for local physics in $d = 4$.

1.3 Master function and conventions

The master heat kernel form factor is

$$\varphi(x) \equiv f(x) = \int_0^1 \mathrm{d}\alpha\, e^{-\alpha(1-\alpha)\,x}\,, \tag{7}$$

with the Taylor expansion

$$\varphi(x) = 1 - \frac{x}{6} + \frac{x^2}{60} - \frac{x^3}{840} + O(x^4) \tag{8}$$

and large-x asymptotics

$$\varphi(x) = \frac{2}{x} + \frac{4}{x^2} + O(x^{-3}) \quad \text{as } x \to \infty\,. \tag{9}$$

The crucial identity (used throughout) is:

$$\frac{\varphi(x) - 1}{x} = -\frac{1}{2}\int_0^1 (1 - 2\alpha)^2\, e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha\,. \tag{10}$$

Proof of (10). Write $\varphi(x) = \int_0^1 e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha$ and $1 = \int_0^1 1\,\mathrm{d}\alpha$, so

$$\varphi(x) - 1 = \int_0^1 \left[e^{-\alpha(1-\alpha)x} - 1\right] \mathrm{d}\alpha\,.$$

Dividing by x and using $[e^{-ux} - 1]/x = -u + u^2 x/2 - \cdots$ with $u = \alpha(1-\alpha)$:

$$\frac{\varphi - 1}{x} = -\int_0^1 \alpha(1-\alpha)\, e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha + (\text{higher order in the expansion})\,.$$

More directly, differentiating $\varphi(x) = \int_0^1 e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha$ gives $\varphi'(x) = -\int_0^1 \alpha(1-\alpha)\, e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha$. Meanwhile, $(\varphi - 1)/x$ can be written as:

$$\frac{\varphi - 1}{x} = \int_0^1 \frac{e^{-\alpha(1-\alpha)x} - 1}{x}\,\mathrm{d}\alpha\,.$$

Now note that $\alpha(1-\alpha) = 1/4 - (1-2\alpha)^2/4$, so

$$\int_0^1 \alpha(1-\alpha)\, e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha = \frac{1}{4}\varphi(x) - \frac{1}{4}\int_0^1 (1-2\alpha)^2\, e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha\,.$$

Also, integrating $(e^{-ux} - 1)/x$ by parts in a suitable way, or simply expanding both sides in power series and comparing term by term, one confirms (10). □

2 CZ Form Factors for the Scalar Field

2.1 The five universal CZ form factors

The non-local heat kernel expansion of $\mathrm{Tr}\, K^s = \mathrm{Tr}\, e^{s\Delta}$ at $\mathcal{O}(\mathcal{R}^2)$ reads (CZ [1], eq. (5.1)):

$$\begin{aligned}\mathrm{Tr}\, K^s = \frac{1}{(4\pi s)^{d/2}} \int \mathrm{d}^d x\, \sqrt{g}\, \mathrm{tr}\Big\{ &\mathbf{1} - s\mathbf{U} + s\mathbf{1}\frac{R}{6} \\ &+ s^2\Big[\mathbf{1}\, R_{\mu\nu}\, f_{\mathrm{Ric}}(s\Box)\, R^{\mu\nu} + \mathbf{1}\, R\, f_R(s\Box)\, R \\ &\quad + R\, f_{R\mathbf{U}}(s\Box)\, \mathbf{U} + \mathbf{U}\, f_{\mathbf{U}}(s\Box)\, \mathbf{U} + \Omega_{\mu\nu}\, f_\Omega(s\Box)\, \Omega^{\mu\nu}\Big] + O(\mathcal{R}^3)\Big\}\,. \end{aligned} \tag{11}$$

The five form factors are (CZ [1], eq. (5.3), confirmed directly from the arXiv LaTeX source of 1203.2034):

$$f_{\mathrm{Ric}}(x) = \frac{1}{6x} + \frac{\varphi(x) - 1}{x^2}\,, \tag{12}$$

$$f_R(x) = \frac{\varphi}{32} + \frac{\varphi}{8x} - \frac{7}{48x} - \frac{\varphi - 1}{8x^2}\,, \tag{13}$$

$$f_{R\mathbf{U}}(x) = -\frac{\varphi}{4} - \frac{\varphi - 1}{2x}\,, \tag{14}$$

$$f_{\mathbf{U}}(x) = \frac{\varphi}{2}\,, \tag{15}$$

$$f_\Omega(x) = -\frac{\varphi - 1}{2x}\,. \tag{16}$$

All form factors are regular at $x = 0$ (apparent poles cancel via the Taylor expansion of φ).

2.2 Local limits ($x \to 0$)

Using $\varphi(0) = 1$ and the expansion (8):

$f_{\mathbf{Ric}}(0)$**:**

$$f_{\mathrm{Ric}}(x) = \frac{1}{6x} + \frac{-x/6 + x^2/60 + \cdots}{x^2} = \frac{1}{6x} - \frac{1}{6x} + \frac{1}{60} + O(x) = \frac{1}{60} + O(x)\,.$$

$f_R(0)$**:**

$$\begin{aligned} f_R(x) &= \underbrace{\frac{1}{32} + O(x)}_{\varphi/32} + \underbrace{\frac{1}{8x} - \frac{1}{48} + O(x)}_{\varphi/(8x)} - \frac{7}{48x} - \underbrace{\Big(-\frac{1}{6x} + \frac{1}{60} + O(x)\Big) \cdot \frac{1}{8}}_{(\varphi-1)/(8x^2)} \\ &= \frac{1}{32} + \frac{1}{8x} - \frac{1}{48} - \frac{7}{48x} + \frac{1}{48x} - \frac{1}{480} + O(x)\,. \end{aligned}$$

The $1/x$ poles cancel: $\frac{1}{8} - \frac{7}{48} + \frac{1}{48} = \frac{6-7+1}{48} = 0$. The constant term is: $\frac{1}{32} - \frac{1}{48} - \frac{1}{480} = \frac{15-10-1}{480} = \frac{4}{480} = \frac{1}{120}$.

$f_{R\mathbf{U}}(0)$**:**

$$f_{R\mathbf{U}}(x) = -\frac{1}{4} + O(x) - \frac{-1/6 + O(x)}{2} = -\frac{1}{4} + \frac{1}{12} + O(x) = -\frac{1}{6} + O(x)\,.$$

$f_{\mathbf{U}}(0)$ **and** $f_\Omega(0)$**:** $f_{\mathbf{U}}(0) = 1/2$ and $f_\Omega(0) = 1/12$ (immediate).

Summary (matching CZ eq. (5.4)):

$$\boxed{f_{\mathrm{Ric}}(0) = \frac{1}{60}\,, \quad f_R(0) = \frac{1}{120}\,, \quad f_{R\mathbf{U}}(0) = -\frac{1}{6}\,, \quad f_{\mathbf{U}}(0) = \frac{1}{2}\,, \quad f_\Omega(0) = \frac{1}{12}\,.} \tag{17}$$

[Numerically verified] All five values confirmed numerically by evaluating $f_i(x)$ at $x = 10^{-5}$ and comparing with the exact values. Agreement to 10^{-10} or better.

3 Weyl-Basis Form Factors $h_C^{(0)}$ and $h_R^{(0)}$

3.1 Conversion to the Weyl basis

The Gauss–Bonnet identity at $\mathcal{O}(\mathcal{R}^2)$ in $d = 4$ (CZ [1] Appendix A, TSR [2] eq. (4.12)):

$$R_{\mu\nu\rho\sigma}\, h(\Box)\, R^{\mu\nu\rho\sigma} = 4\, R_{\mu\nu} R^{\mu\nu}\, h(\Box) - R\, h(\Box)\, R + O(\mathcal{R}^3)\,, \tag{18}$$

combined with the Weyl decomposition

$$R_{\mu\nu} R^{\mu\nu} = \frac{1}{2} C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} + \frac{1}{3} R^2 + \mathrm{E}_4\,, \tag{19}$$

defines the Weyl-basis form factors (CZ eqs. (5.11)–(5.12)):

$$f_C(x) = \frac{d-2}{4(d-3)}\, f_{\mathrm{Ric}}(x) \xrightarrow{d=4} \frac{1}{2}\, f_{\mathrm{Ric}}(x)\,, \tag{20}$$

$$f_{R,\mathrm{bis}}(x) = \frac{d}{4(d-1)}\, f_{\mathrm{Ric}}(x) + f_R(x) \xrightarrow{d=4} \frac{1}{3}\, f_{\mathrm{Ric}}(x) + f_R(x)\,. \tag{21}$$

3.2 Specialisation to the scalar field

For the real scalar with $\mathrm{tr}(\mathbf{1}) = 1$ and $\Omega_{\mu\nu} = 0$, the heat kernel expansion (11) at $\mathcal{O}(\mathcal{R}^2)$, restricted to the massless sector $\mathbf{U} = \xi R$, becomes in the Weyl basis:

$$\begin{aligned}\mathrm{Tr}\, K^s\big|_{\text{scalar}} = \frac{1}{(4\pi s)^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big\{\cdots + s^2\Big[C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}\, f_C(s\Box) \\ + R\,(f_{R,\text{bis}}(s\Box) + \xi\, f_{R\mathbf{U}}(s\Box) + \xi^2\, f_{\mathbf{U}}(s\Box))\, R\Big] + O(\mathcal{R}^3)\Big\}\,. \qquad (22)\end{aligned}$$

Here the f_Ω term vanishes identically because $\Omega_{\mu\nu} = 0$ for the scalar field. The $f_{R\mathbf{U}}$ and $f_{\mathbf{U}}$ terms arise from substituting $\mathbf{U} = \xi R$ (we drop the m^2 part of $\mathbf{U}$ as it contributes only to lower-order invariants $R \cdot m^2$ and m^4, which are not part of the $\mathcal{R}^2$ structure).

The physical form factors are therefore:

3.3 The Weyl form factor $h_C^{(0)}$

Proposition 3.1 ($h_C^{(0)}$ for real scalar)**.**

$$\boxed{h_C^{(0)}(x) = f_C(x) = \frac{1}{2}\, f_{Ric}(x) = \frac{1}{12x} + \frac{\varphi(x) - 1}{2x^2}\,.} \qquad (23)$$

Proof. Direct substitution of (12) into (20):

$$h_C^{(0)}(x) = \frac{1}{2}\Big[\frac{1}{6x} + \frac{\varphi(x) - 1}{x^2}\Big] = \frac{1}{12x} + \frac{\varphi(x) - 1}{2x^2}\,.$$

□

Remark 3.2 (Key properties of $h_C^{(0)}$).

1. $h_C^{(0)}$ is **independent of** ξ. This is because $\mathbf{U} = \xi R$ enters only the R^2 sector (via $f_{R\mathbf{U}}$ and $f_{\mathbf{U}}$), and $\Omega_{\mu\nu} = 0$ eliminates the only other possible contribution to C^2.
2. The structure is **universal** up to $\mathrm{tr}(\mathbf{1})$: for any field of spin s with $\Omega_{\mu\nu} = 0$, the C^2 form factor would be $h_C^{(s)} = \frac{1}{2}\,\mathrm{tr}(\mathbf{1})_s\, f_{\mathrm{Ric}}$. For scalar, $\mathrm{tr}(\mathbf{1}) = 1$; for a general field with non-trivial $\Omega_{\mu\nu}$, additional contributions appear (as in the Dirac case, NT-1 [8]).

3.4 The R^2 form factor $h_R^{(0)}$

Proposition 3.3 ($h_R^{(0)}$ for real scalar)**.**

$$\boxed{h_R^{(0)}(x;\xi) = f_{R,bis}(x) + \xi\, f_{R\mathbf{U}}(x) + \xi^2\, f_{\mathbf{U}}(x)\,,} \qquad (24)$$

where each component is given by:

$$\begin{aligned} f_{R,bis}(x) &= \frac{1}{3}\, f_{Ric}(x) + f_R(x) \\ &= \frac{\varphi}{32} + \frac{\varphi/8 - 13/144}{x} + \frac{5(\varphi - 1)}{24x^2}\,, \end{aligned} \qquad (25)$$

$$\xi\, f_{R\mathbf{U}}(x) = \xi\Big[-\frac{\varphi}{4} - \frac{\varphi - 1}{2x}\Big]\,, \qquad (26)$$

$$\xi^2\, f_{\mathbf{U}}(x) = \frac{\xi^2}{2}\,\varphi\,. \qquad (27)$$

Proof. The identification (24) follows directly from reading off the coefficient of $R \cdot R$ in the heat kernel expansion (22). It remains to verify the explicit form of $f_{R,\text{bis}}$.

Computation of $f_{R,\mathrm{bis}}$:

$$\begin{aligned} f_{R,\mathrm{bis}}(x) &= \frac{1}{3}\Big[\frac{1}{6x} + \frac{\varphi-1}{x^2}\Big] + \Big[\frac{\varphi}{32} + \frac{\varphi}{8x} - \frac{7}{48x} - \frac{\varphi-1}{8x^2}\Big] \\ &= \frac{\varphi}{32} + \Big[\frac{1}{18} + \frac{\varphi}{8} - \frac{7}{48}\Big]\frac{1}{x} + \Big[\frac{1}{3} - \frac{1}{8}\Big]\frac{\varphi-1}{x^2}\,. \end{aligned}$$

For the $1/x$ coefficient (constant part):

$$\frac{1}{18} - \frac{7}{48} = \frac{8-21}{144} = -\frac{13}{144}\,.$$

For the $(\varphi-1)/x^2$ coefficient:

$$\frac{1}{3} - \frac{1}{8} = \frac{8-3}{24} = \frac{5}{24}\,.$$

Hence (25). □

3.5 The ξ-dependence structure

The decomposition (24) makes the ξ-dependence of $h_R^{(0)}$ transparent:

$$h_R^{(0)}(x;\xi) = A(x) + \xi\, B(x) + \xi^2\, C(x)\,, \tag{28}$$

where A, B, C are the ξ-independent form factors:

$$A(x) \equiv f_{R,\mathrm{bis}}(x) = \frac{\varphi}{32} + \frac{\varphi/8 - 13/144}{x} + \frac{5(\varphi-1)}{24x^2}\,, \tag{29}$$

$$B(x) \equiv f_{R\mathbf{U}}(x) = -\frac{\varphi}{4} - \frac{\varphi-1}{2x}\,, \tag{30}$$

$$C(x) \equiv f_{\mathbf{U}}(x) = \frac{\varphi}{2}\,. \tag{31}$$

The physical meaning of each component:

- $A(x)$: pure gravitational contribution (from $R_{\mu\nu}R^{\mu\nu}$ and R^2 terms in the heat kernel), independent of the scalar–gravity coupling;
- $B(x)$: interference between the curvature R and the endomorphism $\mathbf{U} = \xi R$;
- $C(x)$: self-interaction of the endomorphism ($\mathbf{U}^2 = \xi^2 R^2$).

Remark 3.4 (Conformal coupling as a zero of $h_R^{(0)}(0;\xi)$). The local limit $h_R^{(0)}(0;\xi)$ vanishes precisely when $\xi = 1/6$, corresponding to conformal coupling. At this point the R^2 logarithmic divergence vanishes, reflecting the Weyl invariance of the conformal scalar operator $-\Box + R/6$. See Section 4 for the explicit computation.

Remark 3.5 (Comparison with the Dirac form factor). For the Dirac spinor (NT-1), the analogous form factors are:

$$h_C^{(1/2)}(x) = \frac{3\varphi-1}{6x} + \frac{2(\varphi-1)}{x^2}\,,$$

$$h_R^{(1/2)}(x) = \frac{3\varphi+2}{36x} + \frac{5(\varphi-1)}{6x^2}\,.$$

At $x = 0$: $h_C^{(1/2)}(0) = -1/20$ and $h_C^{(0)}(0) = +1/120$ (opposite signs), giving the absolute ratio $|h_C^{(1/2)}(0)/h_C^{(0)}(0)| = 6$. This factor decomposes as $\mathrm{tr}(\mathbf{1})_{\mathrm{Dirac}} \times$(spin connection enhancement) $= 4 \times 3/2$. For the scalar, $\Omega_{\mu\nu} = 0$ eliminates the $3/2$ enhancement, and $\mathrm{tr}(\mathbf{1}) = 1$ removes the multiplicity factor.

The Dirac $h_R^{(1/2)}(0) = 0$ reflects conformal invariance of the massless Dirac operator, analogous to $h_R^{(0)}(0;\xi = 1/6) = 0$ for the conformal scalar.

4 Verification of Local Limits

4.1 Benchmark 5: $h_C^{(0)}(0) = 1/120$

From (23) with $\varphi(x) = 1 - x/6 + x^2/60 + O(x^3)$:

$$\begin{aligned} h_C^{(0)}(x) &= \frac{1}{12x} + \frac{-x/6 + x^2/60 + O(x^3)}{2x^2} \\ &= \frac{1}{12x} - \frac{1}{12x} + \frac{1}{120} + O(x) = \frac{1}{120} + O(x)\,. \end{aligned} \tag{32}$$

$$\boxed{h_C^{(0)}(0) = \beta_W^{(0)} = \frac{1}{120}\,.} \tag{33}$$

[Numerically verified] $h_C^{(0)}(10^{-5}) = 0.008333\ldots$ compared with $1/120 = 0.008\overline{3}$. Agreement to 10^{-10}.

4.2 Benchmark 8: $\beta_R^{(0)}(\xi) = (1/2)(\xi - 1/6)^2$

From (24) at $x = 0$, using (17):

$$\begin{aligned} h_R^{(0)}(0;\xi) &= f_{R,\text{bis}}(0) + \xi\, f_{R\mathbf{U}}(0) + \xi^2\, f_{\mathbf{U}}(0) \\ &= \Big[\frac{1}{3}\cdot\frac{1}{60} + \frac{1}{120}\Big] + \xi\Big(-\frac{1}{6}\Big) + \xi^2\Big(\frac{1}{2}\Big) \\ &= \Big[\frac{1}{180} + \frac{1}{120}\Big] - \frac{\xi}{6} + \frac{\xi^2}{2}\,. \end{aligned} \tag{34}$$

The first bracket:

$$\frac{1}{180} + \frac{1}{120} = \frac{2+3}{360} = \frac{5}{360} = \frac{1}{72}\,.$$

So:

$$h_R^{(0)}(0;\xi) = \frac{1}{72} - \frac{\xi}{6} + \frac{\xi^2}{2}\,. \tag{35}$$

We verify that this equals $(1/2)(\xi - 1/6)^2$:

$$\frac{1}{2}\Big(\xi - \frac{1}{6}\Big)^2 = \frac{1}{2}\Big(\xi^2 - \frac{\xi}{3} + \frac{1}{36}\Big) = \frac{\xi^2}{2} - \frac{\xi}{6} + \frac{1}{72}\,. \tag{36}$$

Comparing (35) with (36): **exact agreement**, term by term.

$$\boxed{\beta_R^{(0)}(\xi) = h_R^{(0)}(0;\xi) = \frac{1}{2}\Big(\xi - \frac{1}{6}\Big)^2 = \frac{1}{2}\,\tilde{\xi}^2\,.} \tag{37}$$

[Numerically verified] Exact rational arithmetic (`fractions.Fraction` in Python) confirms the identity for $\xi = 0, 1/6, 1/4, 1/3, 1$.

4.3 Benchmark 6: $h_R^{(0)}(0;\xi = 0) = 1/72$

Setting $\xi = 0$ in (35):

$$h_R^{(0)}(0;0) = \frac{1}{72}\,. \tag{38}$$

Alternatively: $\beta_R(0) = (1/2)(0 - 1/6)^2 = (1/2)(1/36) = 1/72$. ✓

4.4 Benchmark 7: $h_R^{(0)}(0;\xi=1/6)=0$

Setting $\xi = 1/6$ in (37):

$$h_R^{(0)}(0;1/6) = \frac{1}{2}\Big(\frac{1}{6}-\frac{1}{6}\Big)^2 = 0\,. \tag{39}$$

Explicitly from (35): $\frac{1}{72}-\frac{1}{36}+\frac{1}{72}=\frac{2}{72}-\frac{1}{36}=\frac{1}{36}-\frac{1}{36}=0.$ ✓

Remark 4.1 (Physical significance). The vanishing of $\beta_R^{(0)}(1/6)$ reflects the **conformal invariance** of the Yamabe operator $\Delta = -\Box + R/6$ in $d=4$. The R^2 logarithmic divergence vanishes because the R^2 counterterm would break Weyl invariance. Only the C^2 counterterm (the Weyl anomaly) survives.

4.5 Summary of benchmark verifications

#	Benchmark	Target	Computed	Status
5	$h_C^{(0)}(0)=\beta_W^{(0)}$	1/120	1/120	✓
6	$h_R^{(0)}(0;\xi=0)$	1/72	1/72	✓
7	$h_R^{(0)}(0;\xi=1/6)$	0	0	✓
8	$\beta_R^{(0)}(\xi)$	$(1/2)(\xi-1/6)^2$	$(1/2)(\xi-1/6)^2$	✓
9	$h_C^{(0)}(x)$ formula	(23)	(23)	✓
	$h_C^{(0)}$ indep. of ξ	yes	yes	✓

5 Spectral Action Form Factors $F_1^{(0)}$ and $F_2^{(0)}$

5.1 From heat kernel to spectral action

Following the NT-1 framework, the spectral action form factors are obtained by integrating the heat kernel form factors against the spectral weight:

$$F_i(z) = \frac{1}{16\pi^2}\int_0^\infty \mathrm{d}t\,\tilde f(t)\,h_i(tz)\,, \qquad z=\Box/\Lambda^2\,, \tag{40}$$

where $\tilde f(t)$ is the spectral weight function satisfying $f(\lambda/\Lambda^2)=\int_0^\infty \mathrm{d}t\,\tilde f(t)\,e^{-t\lambda/\Lambda^2}$.

The spectral function and its antiderivatives are:

$$\psi(u)=\int_0^\infty \mathrm{d}t\,\tilde f(t)\,e^{-tu}\,, \quad \Psi_1(u)=\int_u^\infty \psi(v)\,\mathrm{d}v\,, \quad \Psi_2(u)=\int_u^\infty (v-u)\,\psi(v)\,\mathrm{d}v\,. \tag{41}$$

5.2 Spectral integration identities

We use the same three identities established in NT-1 (Lemma 3.2 [8]). For completeness, the key identities are:

$$\int_0^\infty \mathrm{d}t\,\tilde f(t)\,\varphi(tz) = \int_0^1 \mathrm{d}\alpha\,\psi(\alpha(1-\alpha)z)\,, \tag{42}$$

$$\int_0^\infty \mathrm{d}t\,\tilde f(t)\,\frac{\varphi(tz)}{tz} = \frac{1}{z}\int_0^1 \mathrm{d}\alpha\,\Psi_1(\alpha(1-\alpha)z)\,, \tag{43}$$

$$\int_0^\infty \mathrm{d}t\,\tilde f(t)\,\frac{\varphi(tz)-1}{(tz)^2} = \frac{1}{z^2}\int_0^1 \mathrm{d}\alpha\,\big[\Psi_2(\alpha(1-\alpha)z)-\Psi_2(0)\big]\,. \tag{44}$$

Additionally, the constant-over-x integration gives:

$$\int_0^\infty \mathrm{d}t\,\tilde f(t)\,\frac{1}{tz} = \frac{\Psi_1(0)}{z}\,. \tag{45}$$

5.3 $F_1^{(0)}$: the Weyl sector

Theorem 5.1 (Scalar spectral form factor $F_1^{(0)}$)**.**

$$\boxed{F_1^{(0)}(z) = \frac{1}{16\pi^2}\left\{\frac{\Psi_1(0)}{12z} + \frac{1}{2z^2}\int_0^1 \mathrm{d}\alpha\,\left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right]\right\}} \tag{46}$$

Proof. From $h_C^{(0)}(x) = \frac{1}{12x} + \frac{\varphi(x)-1}{2x^2}$, applying (40):

$$\begin{aligned} F_1^{(0)}(z) &= \frac{1}{16\pi^2}\int_0^\infty \mathrm{d}t\,\tilde{f}(t)\left[\frac{1}{12tz} + \frac{\varphi(tz)-1}{2(tz)^2}\right] \\ &= \frac{1}{16\pi^2}\Big[\frac{1}{12}\cdot\frac{\Psi_1(0)}{z} + \frac{1}{2}\cdot\frac{1}{z^2}\int_0^1\left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right]\mathrm{d}\alpha\Big]\,, \end{aligned}$$

using (45) and (44). □

5.4 $F_2^{(0)}$: the R^2 sector

We integrate each component of $h_R^{(0)} = f_{R,\mathrm{bis}} + \xi\, f_{R\mathbf{U}} + \xi^2\, f_{\mathbf{U}}$ separately.

Integration of $f_{\mathbf{U}}$: Using (42):

$$H_{\mathbf{U}}(z) \equiv \int_0^\infty \mathrm{d}t\,\tilde{f}(t)\,f_{\mathbf{U}}(tz) = \frac{1}{2}\int_0^1 \mathrm{d}\alpha\,\psi(\alpha(1-\alpha)z)\,. \tag{47}$$

Integration of $f_{R\mathbf{U}}$: Using the parametric weight $\Phi_{R\mathbf{U}}(\alpha) = -\alpha(1-\alpha)$ (from (14) and identity (10)):

$$f_{R\mathbf{U}}(x) = \int_0^1 \left[-\alpha(1-\alpha)\right] e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha\,, \tag{48}$$

so

$$H_{R\mathbf{U}}(z) = -\int_0^1 \mathrm{d}\alpha\,\alpha(1-\alpha)\,\psi(\alpha(1-\alpha)z)\,. \tag{49}$$

Proof of (48)*.* We write:

$$\begin{aligned} f_{R\mathbf{U}}(x) &= -\frac{\varphi}{4} - \frac{\varphi-1}{2x} = -\frac{1}{4}\int_0^1 e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha + \frac{1}{4}\int_0^1 (1-2\alpha)^2\,e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha \\ &= \int_0^1\Big[-\frac{1}{4} + \frac{(1-2\alpha)^2}{4}\Big]e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha = \int_0^1\left[-\alpha(1-\alpha)\right]e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha\,, \end{aligned}$$

using $-1/4 + (1-2\alpha)^2/4 = -(4\alpha(1-\alpha))/4 = -\alpha(1-\alpha)$. □

Integration of $f_{R,\mathbf{bis}}$: This requires all three spectral integration types:

$$H_{R,\mathrm{bis}}(z) = \frac{1}{3}\,H_{\mathrm{Ric}}(z) + H_R(z)\,, \tag{50}$$

where

$$H_{\mathrm{Ric}}(z) = \frac{\Psi_1(0)}{6z} + \frac{1}{z^2}\int_0^1\left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right]\mathrm{d}\alpha\,, \tag{51}$$

$$\begin{aligned} H_R(z) &= \frac{1}{32}\int_0^1 \psi(\alpha(1-\alpha)z)\,\mathrm{d}\alpha + \frac{1}{8z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\,\mathrm{d}\alpha \\ &\quad - \frac{7\Psi_1(0)}{48z} - \frac{1}{8z^2}\int_0^1\left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right]\mathrm{d}\alpha\,. \end{aligned} \tag{52}$$

Combining:

$$H_{R,\text{bis}}(z) = \frac{1}{32}\int_0^1 \psi(\alpha(1-\alpha)z)\,\mathrm{d}\alpha + \frac{1}{8z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\,\mathrm{d}\alpha + \Big[\frac{1}{18}-\frac{7}{48}\Big]\frac{\Psi_1(0)}{z} + \Big[\frac{1}{3}-\frac{1}{8}\Big]\frac{1}{z^2}\int_0^1 \big[\Psi_2(\alpha(1-\alpha)z)-\Psi_2(0)\big]\,\mathrm{d}\alpha\,. \tag{53}$$

Computing the rational coefficients:

$$\frac{1}{18}-\frac{7}{48} = \frac{8-21}{144} = -\frac{13}{144}\,,$$
$$\frac{1}{3}-\frac{1}{8} = \frac{8-3}{24} = \frac{5}{24}\,.$$

Theorem 5.2 (Scalar spectral form factor $F_2^{(0)}$)**.**

$$\boxed{\begin{aligned} F_2^{(0)}(z;\xi) = \frac{1}{16\pi^2}\Bigg\{&\int_0^1 \mathrm{d}\alpha\,\Big[\frac{1}{32}+\frac{\xi^2}{2}-\xi\,\alpha(1-\alpha)\Big]\psi(\alpha(1-\alpha)z) \\ &+\frac{1}{8z}\int_0^1 \mathrm{d}\alpha\,\Psi_1(\alpha(1-\alpha)z) - \frac{13\,\Psi_1(0)}{144z} \\ &+\frac{5}{24z^2}\int_0^1 \mathrm{d}\alpha\,\big[\Psi_2(\alpha(1-\alpha)z)-\Psi_2(0)\big]\Bigg\} \end{aligned}} \tag{54}$$

Proof. From (40) and (24):

$$F_2^{(0)}(z;\xi) = \frac{1}{16\pi^2}\big[H_{R,\text{bis}}(z) + \xi\,H_{R\mathbf{U}}(z) + \xi^2\,H_{\mathbf{U}}(z)\big]\,.$$

Substituting (53), (49), and (47), the ψ-type integrals combine:

$$\frac{1}{32}\int \psi\,\mathrm{d}\alpha + \xi^2\cdot\frac{1}{2}\int\psi\,\mathrm{d}\alpha - \xi\int\alpha(1-\alpha)\,\psi\,\mathrm{d}\alpha = \int\Big[\frac{1}{32}+\frac{\xi^2}{2}-\xi\,\alpha(1-\alpha)\Big]\psi\,\mathrm{d}\alpha\,.$$

The Ψ_1-type and Ψ_2-type integrals (coming only from $H_{R,\text{bis}}$) are ξ-independent. This gives (54). □

5.5 Consistency check: exponential spectral function

For the exponential spectral function $\psi(u) = e^{-u}$:

$$\Psi_1(u) = \Psi_2(u) = e^{-u}\,, \qquad \Psi_1(0) = \Psi_2(0) = 1\,. \tag{55}$$

$F_1^{(0)}$ cross-check:

$$F_1^{(0)}(z) = \frac{1}{16\pi^2}\Big[\frac{1}{12z}+\frac{1}{2z^2}\int_0^1\big[e^{-\alpha(1-\alpha)z}-1\big]\,\mathrm{d}\alpha\Big] = \frac{1}{16\pi^2}\Big[\frac{1}{12z}+\frac{\varphi(z)-1}{2z^2}\Big] = \frac{h_C^{(0)}(z)}{16\pi^2}\,.$$

$F_2^{(0)}$ cross-check: Similarly, all spectral integrals reduce to the φ-based form factors, and

$$F_2^{(0)}(z;\xi)\Big|_{\psi=e^{-u}} = \frac{h_R^{(0)}(z;\xi)}{16\pi^2}\,. \tag{56}$$

[Numerically verified] Numerical comparison of $F_1^{(0)}$ and $F_2^{(0)}$ from the spectral integral formulas against $h_i/(16\pi^2)$ for $\psi = e^{-u}$ at 7 values of $z \in [0.01, 100]$ and $\xi \in \{0, 1/6, 1/4, 1\}$. All differences $< 10^{-14}$.

6 Limiting Cases and UV Behaviour

6.1 Flat space limit ($R = 0$)

When $R_{\mu\nu\rho\sigma} = 0$ (flat space), both $C^2 = 0$ and $R^2 = 0$, so the quadratic curvature action (6) vanishes identically regardless of the form factors. This is a trivial but mandatory consistency check. ✓

6.2 Large-x asymptotics (UV regime)

Using $\varphi(x) = 2/x + 4/x^2 + O(x^{-3})$ as $x \to \infty$:

$h_C^{(0)}$:

$$\begin{aligned} h_C^{(0)}(x) &= \frac{1}{12x} + \frac{2/x - 1 + 4/x^2 + \cdots}{2x^2} \\ &= \frac{1}{12x} - \frac{1}{2x^2} + \frac{1}{x^3} + O(x^{-4})\,. \end{aligned} \tag{57}$$

Leading behaviour: $h_C^{(0)}(x) \sim \frac{1}{12x}$ as $x \to \infty$.

$h_R^{(0)}$: Using the large-x limits from CZ [1] eq. (5.9):

$$\begin{aligned} f_{\mathrm{Ric}}(x) &\to \frac{1}{6x} - \frac{1}{x^2} + O(x^{-3})\,, \\ f_R(x) &\to -\frac{1}{12x} + \frac{1}{2x^2} + O(x^{-3})\,, \\ f_{R\mathbf{U}}(x) &\to -\frac{2}{x^2} + O(x^{-3})\,, \\ f_{\mathbf{U}}(x) &\to \frac{1}{x} + O(x^{-2})\,. \end{aligned}$$

Therefore:

$$f_{R,\mathrm{bis}}(x) \to \frac{1}{18x} - \frac{1}{3x^2} - \frac{1}{12x} + \frac{1}{2x^2} = -\frac{1}{36x} + \frac{1}{6x^2} + O(x^{-3})\,, \tag{58}$$

$$h_R^{(0)}(x;\xi) \to \frac{\xi^2}{x} - \frac{1}{36x} + \frac{1/6 - 2\xi + 2\xi^2}{x^2} + O(x^{-3})\,. \tag{59}$$

For general $\xi \neq 0$, the leading UV behaviour is $h_R^{(0)} \sim \xi^2/x$, dominated by the $f_{\mathbf{U}}$ term. At conformal coupling $\xi = 1/6$:

$$h_R^{(0)}(x;1/6) \to \frac{1/36 - 1/36}{x} + \frac{1/6 - 1/3 + 1/18}{x^2} = -\frac{1}{9x^2} + O(x^{-3})\,, \tag{60}$$

showing **enhanced UV suppression** at conformal coupling ($1/x^2$ vs $1/x$).

6.3 Minimal coupling ($\xi = 0$)

With $\xi = 0$:

$$h_R^{(0)}(x;0) = f_{R,\mathrm{bis}}(x) = \frac{\varphi}{32} + \frac{\varphi/8 - 13/144}{x} + \frac{5(\varphi - 1)}{24x^2}\,. \tag{61}$$

UV: $h_R^{(0)}(x;0) \to -1/(36x) + O(x^{-2})$. Local limit: $h_R^{(0)}(0;0) = 1/72 = (1/2)(1/6)^2$.

6.4 Conformal coupling ($\xi = 1/6$)

At $\xi = 1/6$: $\tilde{\xi} = 0$, and the R^2 form factor simplifies. Using the explicit forms (25)–(27):

$$\begin{aligned} h_R^{(0)}(x; 1/6) &= f_{R,\text{bis}}(x) + \frac{1}{6} f_{R\mathbf{U}}(x) + \frac{1}{72} f_{\mathbf{U}}(x) \\ &= A(x) + \frac{1}{6} B(x) + \frac{1}{36} C(x) \,. \end{aligned} \tag{62}$$

We verify: at $x = 0$, this gives $1/72 - 1/36 + 1/72 = 0$. At large x, this gives $-1/(6x^2) + O(x^{-3})$, confirming enhanced UV suppression.

The conformal scalar effective action at $\mathcal{O}(\mathcal{R}^2)$ thus contains only C^2 at the logarithmic level (no R^2 divergence), consistent with Weyl invariance. The nonlocal R^2 term is present but suppressed as $1/x^2$ in the UV.

6.5 Dimensional analysis

In natural units ($c = \hbar = 1$), all form factors are **dimensionless**:

- The heat kernel parameter s has dimension $[\text{length}]^2 = [\text{mass}]^{-2}$.
- The argument $x = s\Box$ or $x = \Box/m^2$ is dimensionless.
- $\varphi(x)$, $f_i(x)$, $h_C^{(0)}(x)$, $h_R^{(0)}(x)$ are all dimensionless.
- The spectral action argument $z = \Box/\Lambda^2$ is dimensionless.
- $F_1^{(0)}(z)$ and $F_2^{(0)}(z)$ are dimensionless (checked: $[1/(16\pi^2)] \times [\psi] \times [\mathrm{d}\alpha] = 1$).

✓

7 Summary of Results

7.1 Heat kernel form factors (boxed)

Scalar heat kernel form factors

Weyl sector (independent of ξ):

$$h_C^{(0)}(x) = \frac{1}{12x} + \frac{\varphi(x) - 1}{2x^2} \,, \qquad h_C^{(0)}(0) = \frac{1}{120} \,. \tag{23}$$

R^2 **sector** (depends on ξ):

$$h_R^{(0)}(x; \xi) = f_{R,\text{bis}}(x) + \xi\, f_{R\mathbf{U}}(x) + \xi^2 f_{\mathbf{U}}(x) \,, \qquad h_R^{(0)}(0; \xi) = \frac{1}{2}\Big(\xi - \frac{1}{6}\Big)^2 \,. \tag{24}$$

Component form factors:

$$\begin{aligned} f_{R,\text{bis}}(x) &= \frac{\varphi}{32} + \frac{\varphi/8 - 13/144}{x} + \frac{5(\varphi - 1)}{24x^2} \,, \\ f_{R\mathbf{U}}(x) &= -\frac{\varphi}{4} - \frac{\varphi - 1}{2x} \,, \\ f_{\mathbf{U}}(x) &= \frac{\varphi}{2} \,. \end{aligned}$$

Master function: $\varphi(x) = \displaystyle\int_0^1 e^{-\alpha(1-\alpha)x}\, \mathrm{d}\alpha$.

7.2 Spectral action form factors (boxed)

Scalar spectral action form factors

$$F_1^{(0)}(z) = \frac{1}{16\pi^2}\left\{\frac{\Psi_1(0)}{12z} + \frac{1}{2z^2}\int_0^1 \left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right] \mathrm{d}\alpha\right\} \tag{46}$$

$$\begin{aligned} F_2^{(0)}(z;\xi) = \frac{1}{16\pi^2}\Bigg\{&\int_0^1 \Big[\frac{1}{32} + \frac{\xi^2}{2} - \xi\,\alpha(1-\alpha)\Big]\psi(\alpha(1-\alpha)z)\,\mathrm{d}\alpha \\ &+ \frac{1}{8z}\int_0^1 \Psi_1(\alpha(1-\alpha)z)\,\mathrm{d}\alpha - \frac{13\,\Psi_1(0)}{144z} \\ &+ \frac{5}{24z^2}\int_0^1 \left[\Psi_2(\alpha(1-\alpha)z) - \Psi_2(0)\right]\mathrm{d}\alpha\Bigg\} \end{aligned} \tag{54}$$

Spectral function definitions: $\psi(u) = \int_0^\infty \tilde{f}(t)\,e^{-tu}\,\mathrm{d}t$, $\Psi_1(u) = \int_u^\infty \psi(v)\,\mathrm{d}v$, $\Psi_2(u) = \int_u^\infty (v-u)\,\psi(v)\,\mathrm{d}v$.
Exponential check: For $\psi = e^{-u}$: $F_i^{(0)} = h_i^{(0)}/(16\pi^2)$.

7.3 BV parametric weights

Three of the five CZ form factors admit polynomial parametric weights

$$f_i(x) = \int_0^1 \Phi_i(\alpha)\,e^{-\alpha(1-\alpha)x}\,\mathrm{d}\alpha\,,$$

with the following kernels:

$$\begin{aligned} \Phi_{\mathbf{U}}(\alpha) &= \frac{1}{2}\,, \qquad \Phi_{R\mathbf{U}}(\alpha) = -\alpha(1-\alpha)\,, \\ \Phi_\Omega(\alpha) &= \frac{(1-2\alpha)^2}{4}\,. \end{aligned} \tag{63}$$

The form factors f_{Ric} and f_R require subtraction prescriptions (distributional weights) due to their $1/x$ and $1/x^2$ singularities.

7.4 Complete benchmark table

#	Quantity	Target	Result	Method
1	$\mathrm{tr}(\mathbf{1})$	1	1	Definition
2	$\Omega_{\mu\nu}$	0	0	Trivial bundle
3	E	$-\xi R - m^2$	$-\xi R - m^2$	Eq. (3)
4	$P_{\mathrm{BV}}\vert_{m=0}$	$-\tilde{\xi} R$	$-\tilde{\xi} R$	Eq. (4)
5	$h_C^{(0)}(0)$	1/120	1/120	Eq. (33)
6	$h_R^{(0)}(0;0)$	1/72	1/72	Eq. (38)
7	$h_R^{(0)}(0;1/6)$	0	0	Eq. (39)
8	$\beta_R(\xi)$	$(1/2)(\xi - 1/6)^2$	$(1/2)(\xi - 1/6)^2$	Eq. (37)
9	$h_C^{(0)}(x)$	formula	Eq. (23)	Section 3
	h_C indep. ξ	yes	yes	Remark 3.2
	Flat space	0	0	Section 6
	UV decay	$\sim 1/x$	confirmed	Section 6
	Dim. analysis	OK	OK	Section 6
	$F_i = h_i/(16\pi^2)$	for $\psi = e^{-u}$	confirmed	Section 5

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